From: Dr Mehran Behjati <mehranb@sunway.edu.my>
Sent: Monday, 16 March, 2026 11:57 AM
To: CHETAN RAO SONOO <23048457@imail.sunway.edu.my>
Cc: Prof. Ir. Ts. Yap Kian Meng <kmyap@sunway.edu.my>
Subject: Re: Application for Student Research Assistant – APNIC ISIF Asia 2025 Project

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Hi Chetan,

Good day.

Thanks for taking the time to interview with me last week and for your interest in the RA position at HUMAC Lab.

I'm pleased to inform you that you have been selected for this role, and I would be happy to have you join the project.

About the Project

The project is funded under the **APNIC ISIF Asia 2025 program** and focuses on developing a **LoRa-based hiker tracking and emergency alert system for remote rainforest national parks**. The goal is to improve **hiker safety and digital connectivity in areas where cellular networks are unavailable**.

The system will use **low-power wearable trackers** carried by hikers. These devices will transmit location data through **LoRa nodes** to a central server, where the information will be processed and visualised for park authorities.

Project Objectives

The main objectives of the project are:

• Develop a **low-power tracking system** for hikers in remote areas
• Enable **real-time location monitoring** for park rangers
• Implement **geofencing and emergency (SOS) alerts** to improve response time in case of incidents
• Provide a **cost-effective connectivity solution** that can be replicated in other national parks in the Asia-Pacific region

Your Role – Server Development

Your primary responsibility will be to **design and develop the backend server platform** that processes data from the LoRa network and enables live tracking.

The expected tasks include:

1. Data Ingestion Layer

- Receive data from the LoRa network server (via MQTT or API)
- Decode payloads coming from the tracking devices
- Extract fields such as device ID, timestamp, GPS coordinates, battery status, and SOS flag

2. Backend API Development

- Build REST APIs for the tracking platform
- Manage communication between the database and the monitoring dashboard
- Implement device management and data access endpoints

3. Database Design

- Design and implement the database (likely PostgreSQL)
- Store device information, hiker sessions, location history, and alerts
- Implement spatial queries to support geofencing

4. Real-Time Tracking Engine

- Maintain the latest location for each device
- Support real-time updates to the monitoring interface
- Ensure efficient handling of frequent location updates

5. Geofencing Engine

- Implement geofencing logic using park boundary or trail polygons
- Detect when a hiker leaves a defined safe zone
- Trigger alerts when boundary violations occur

6. Emergency Alert Handling

- Detect SOS signals sent from the wearable tracker
- Immediately flag emergency events
- Push alerts to the monitoring system for park rangers

7. Documentation and Version Control

- Maintain clear technical documentation
- Use Git for version control and structured development

You will be working closely with me and the HUMAC Lab team, and this will also give you exposure to **IoT systems, LoRa networking, and real-world research deployment**.

Availability

Could you please let me know your **earliest availability date to start the project (my preference is ASAP)?** As currently you are undertaking your internship you can work remotely.

Once I have this information, we can proceed with the next steps and arrange a short onboarding meeting.

Best,

Mehran Behjati, PhD
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